

Approximate Phase Rigidity and Arithmetic Quadrature Complexity

Sharp rational-resonance exponents,
log-square phase costs, and non-Hölder recovery

Research manuscript prepared with ChatGPT

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30 September 2026

Abstract

Exact phase averaging for the Rvachev up-function has an arithmetic rigidity: at a rational mesh it forces a finite cyclic symmetry, whereas at an irrational mesh it forces Haar measure. We determine what survives when exactness is replaced by a small uniform error.

Our principal result gives the exact order of the optimal error near every positive rational mesh, for every fixed phase budget and every fixed polynomial degree. Write $M_0 = a/b$ in lowest terms, $d = \nu_2(a)$, and let $N \geq b$. For positive, real signed, and complex mass-one filters with at most N atoms, the optimal error on the first $r + 1$ monomials has order $|M - M_0|^{\max\{0, d+1+\lfloor \log_2(N/b) \rfloor - r\}}$. For $N < b$ the error stays bounded away from zero. These lower bounds allow the nodes and unbounded signed weights to depend on M . A regular polygon attains the order, and its leading response is an explicit rescaled coarse-mesh profile.

For an irrational mesh satisfying a finite-type Diophantine lower bound, the optimal positive N -phase error is $\exp[-\Theta((\log N)^2)]$. Equivalently, accuracy ε requires $\exp[\Theta(\sqrt{\log(1/\varepsilon)})]$ phases. We also obtain a finite-frequency Wasserstein inversion inequality. It yields a matching stretched-logarithmic scale of approximate Haar rigidity, while excluding every local Hölder inverse estimate. Finally, specially constructed Liouville meshes permit arbitrarily small errors along prescribed subsequences and arbitrarily slow inverse moduli. All results have ordinary proofs; the package includes exact arithmetic checks and numerical diagnostics.

Status and scope. Unrefereed research manuscript. The local optimal exponent answers the fixed-budget near-rational question in the inspected ProveIt phase-averaging report. The log-square statements identify the scale, not an optimal leading constant. No new Lean or Rocq verification is claimed, and worldwide novelty has not been certified.

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1 The problem, the repository baseline, and the new results

1.1 Why exact rigidity is not a stability theorem

A phase-averaged lattice quadrature rule can be exact for every overall translation only if its phase measure annihilates the appropriate Fourier aliases. For the Rvachev up-function those aliases have a particularly explicit zero set. The ProveIt reports *Rigidity and Quantitative Limits of Phase Averaging* and *Complete Superconvergence Phases and Optimal Phase Filters* classify exact filters and obtain finite-support obstructions [3, 4]. At an irrational mesh, even uniform exactness on constants forces continuous Haar averaging. At a rational mesh a/b , the symmetry required through polynomial degree r has order

$$L_r = b 2^{\max(0, r - v_2(a))}. \quad (1.1)$$

These are inherited results, not discoveries of this article.

The same rigidity report asks how the quantitative obstruction degenerates as an irrational mesh approaches a rational scale, and how positive and signed filters compare away from exactness. These questions are not answered by a classification at a single parameter value. In particular, an optimizing filter may move its nodes, change its weights, and, when signs are allowed, let those weights become arbitrarily large as the mesh changes.

There is a second gap between exactness and approximation. If Haar measure is the only exact phase measure, does an almost-exact rule have to be close to Haar measure at an effective rate? The answer is yes for each fixed irrational mesh, but never at a Hölder rate. Arithmetic assumptions determine whether one can give a useful uniform scale.

1.2 The three main conclusions

The local result, Theorem 4.1, gives a complete answer at every fixed budget N . The exponent depends only on the reduced denominator of the nearby rational mesh, the two-adic valuation of its numerator, the budget, and the requested degree. Its proof combines a finite Fourier-moment obstruction with orders of vanishing of a finite list of multipliers. Thus it remains valid without any compactness hypothesis on signed weights.

The global irrational result, Theorem 7.1, is different in nature: N tends to infinity at a fixed irrational mesh. Positivity upgrades an exponentially weak moment obstruction to a sharp $1/N$ obstruction. Together with explicit two-sided Fourier estimates, it proves

$$-\log \mathcal{E}_{N,r}^+(M) = \Theta((\log N)^2) \quad (1.2)$$

under a finite-type Diophantine hypothesis. The constants in the Θ notation depend on the fixed mesh, degree, and arithmetic parameters.

The inverse result, Theorem 8.1, bounds the Wasserstein distance of a phase probability measure from Haar measure using finitely many nonzero multipliers. This gives a precise stability mechanism and a precise limitation:

$$\omega_{M,r}(\varepsilon) = \exp[-\Theta(\sqrt{\log(1/\varepsilon)})] \quad (1.3)$$

for finite-type irrational meshes. There is no uniform analogue over arbitrary irrational meshes. Theorem 9.1 constructs examples with arbitrarily slow recovery, despite exact uniqueness at zero error.

1.3 Provenance and the meaning of novelty

The repository inspection was anchored to commit

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The two phase-filter reports above were read together with the introduction of the subsequent fixed-phase maximality report [5]. The latter concerns a different problem: a single selected phase at an integer mesh. We do not repackage its dyadic maximality theorem here.

The Fourier product, compactly supported smooth density, and zero multiplicities are classical; Arias de Reyna gives the relevant constructions and formulas [1]. The annihilating-polynomial obstruction is already in the repository’s rigidity report. The positive-moment argument below is an elementary Fejér-kernel argument, and the inversion mechanism belongs to the general family of Fourier smoothing arguments. Wasserstein inversion in deconvolution also appears in a different, statistical setting in the work of Rousseau and Scricciolo [2]. No claim is made that deconvolution, Fejér kernels, or severe inverse instability are new ideas.

The contributions proposed here are the complete local optimal exponent, its explicit leading-profile realization, the positive-filter phase-cost scale for this kernel under a stated arithmetic assumption, and their combination with a quantified inverse theorem and a Liouville obstruction. The supplied proofs are independent of unverified assertions elsewhere in the repository. The source review was targeted, not an audit of the entire corpus or an exhaustive priority search.

2 Kernel, filters, and the error being optimized

2.1 Normalization

Let $V_1, V_2, \dots$ be independent uniform random variables on $[-1, 1]$ and set $Y = \sum_{j \geq 1} 2^{-j} V_j$. Its law has the normalized Rvachev density $u \in C_c^\infty(\mathbb{R})$, supported on $[-1, 1]$, with $u \geq 0$, u even, and $\int u = 1$. In the convention

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i t x} dx, \quad (2.1)$$

its Fourier transform is

$$H(t) = \widehat{u}(t) = \prod_{j=0}^{\infty} \frac{\sin(\pi t/2^j)}{\pi t/2^j}, \quad (2.2)$$

where a factor at zero is interpreted as one. This agrees with the normalization in [1] and the inspected phase reports.

The product converges locally uniformly in the complex plane: after finitely many factors, the difference of a factor from one is bounded by a constant multiple of 4^{-j} on each compact set. It defines an entire function. Since $u \in C_c^\infty$, every derivative of H is rapidly decreasing on the real axis. These facts also follow directly from the product; a quantitative version is proved in Section 6.

Lemma 2.1 (Zero order and leading derivative). *The real zeros of H are exactly the nonzero integers. If $t_0 = 2^v m \neq 0$ with m odd and $p = v + 1$, then*

$$\text{ord}_{t_0} H = p, \quad H^{(p)}(t_0) = -\frac{p!}{t_0^p} H(m/2) \neq 0. \quad (2.3)$$

Proof. Exactly the factors indexed by $0, \dots, v$ vanish at t_0 . Their linear coefficients in $t - t_0$ are $(-1)^{t_0/2^j} / t_0$. The product of these signs is $(-1)^{m(2^{v+1}-1)} = -1$. The remaining product at t_0 is $H(m/2)$, which is nonzero: no factor there vanishes, and the tail is an absolutely convergent product of factors tending to one. This proves the formula. The same nonvanishing-product argument excludes any other real zero. $\square$

Editorial note (ProveIt, 2026-09-30). Lemma 2.1 is machine-checked in the repository, for the same product (2.2) (Fabius.rvachevFourierProduct) and even over $\mathbb{C}$, although this article does not cite it: the complex zeros of H are exactly the nonzero integers (Fabius.rvachevFourierProduct

`_eq_zero_iff`, `Analysis/FabiusFunction/Lean/FabiusFunction/FourierProduct.lean`); at a nonzero integer t_0 the analytic order is $\nu_2(|t_0|) + 1$ (`Fabius.analyticOrderAt_rvachevFourierProduct_int`), and the derivative of that order is $-p! H(m/2)/t_0^p$ with m the odd part of $|t_0|$, which is nonzero (`Fabius.iteratedDeriv_rvachevFourierProduct_int` and `Fabius.iteratedDeriv_rvachevFourierProduct_int_ne_zero`), all in `Analysis/FabiusFunction/Lean/FabiusFunction/IntegerZeroAnalyticOrder.lean`. The zero-multiplicity and leading-derivative half of the “Kernel jets” layer in Section 11 is therefore already formalized. No theorem of this article has a Lean statement.

2.2 The translation-uniform error

Write λ for normalized Haar measure on $\mathbb{R}/\mathbb{Z} = \mathbb{R}/\mathbb{Z}$, and $m_j = \int x^j u(x) dx$. For $M > 0$ define

$$Q_{M,\theta}(P) = \frac{1}{M} \sum_{k \in \mathbb{Z}} P\left(\frac{k+\theta}{M}\right) u\left(\frac{k+\theta}{M}\right). \quad (2.4)$$

The physical spacing is $1/M$. This is a finite sum, because u is compactly supported. A phase filter is a finite complex Borel measure ν on $\mathbb{R}/\mathbb{Z}$ of mass one. We use the convolution convention

$$Q_{M,\nu,\theta}(P) = \int_{\mathbb{R}} Q_{M,\theta-\phi}(P) d\nu(\phi), \quad (2.5)$$

$$D_j(M, \nu; \theta) = Q_{M,\nu,\theta}(x^j) - m_j, \quad E_{M,r}(\nu) = \max_{0 \leq j \leq r} \|D_j(M, \nu; \cdot)\|_{\infty}. \quad (2.6)$$

Using $\theta + \phi$ instead merely reflects the phase measure and does not change any optimized quantity. The norm in (2.6) is also the operator norm for polynomials $\sum_{j=0}^r c_j x^j$ with $\sum_j |c_j| \leq 1$, taking the supremum over θ .

Let $\mathcal{A}_N^+$, $\mathcal{A}_N^{\mathbb{R}}$, and $\mathcal{A}_N^{\mathbb{C}}$ denote, respectively, the positive, real signed, and complex mass-one filters with at most N distinct nonzero atoms. Define

$$\mathcal{E}_{N,r}^{\mathbb{K}}(M) = \inf_{\nu \in \mathcal{A}_N^{\mathbb{K}}} E_{M,r}(\nu), \quad \mathbb{K} \in \{+, \mathbb{R}, \mathbb{C}\}. \quad (2.7)$$

Duplicate phases are coalesced; zero weights do not count. In particular,

$$\mathcal{E}_{N,r}^{\mathbb{C}} \leq \mathcal{E}_{N,r}^{\mathbb{R}} \leq \mathcal{E}_{N,r}^+ \quad (2.8)$$

There is no bound on total variation in either signed class.

Lemma 2.2 (Alias identity and smooth parameter dependence). *Put $\widehat{\nu}(n) = \int e^{-2\pi i n \phi} d\nu(\phi)$. Then*

$$D_j(M, \nu; \theta) = \sum_{n \neq 0} \left(\frac{i}{2\pi}\right)^j H^{(j)}(nM) \widehat{\nu}(n) e^{2\pi i n \theta}. \quad (2.9)$$

For a fixed finite measure, the series and every fixed number of its M - and θ -derivatives converge uniformly when M ranges over a compact subinterval of $(0, \infty)$.

Proof. The Fourier transform of $x^j u(x)$ is $(i/(2\pi))^j H^{(j)}$. Poisson summation for this smooth compactly supported function gives the unfiltered identity. Its zero frequency is m_j . Integrating against ν gives (2.9). A derivative contributes only a fixed power of n and a higher derivative of H . Rapid decrease, together with $|\widehat{\nu}(n)| \leq \|\nu\|_{TV}$, proves uniform convergence. $\square$

2.3 Regular polygons and exact refinement

Let

$$\nu_L = \frac{1}{L} \sum_{h=0}^{L-1} \delta_{h/L}. \quad (2.10)$$

Its Fourier coefficients are one on $L\mathbb{Z}$ and zero elsewhere. Reindexing (2.4), or using (2.9), gives

$$Q_{M,\nu_L,\theta}(P) = Q_{LM,L\theta}(P). \quad (2.11)$$

Thus L equally weighted phase shifts are exactly an L -fold mesh refinement, not a new choice of arbitrary physical weights.

For completeness, Lemmas 2.1 and 2.2 recover the inherited exact classification. At $M = a/b$ in lowest terms, all multipliers through degree r vanish at frequency $n \neq 0$ precisely when $L_r \mid n$, with L_r as in (1.1). Therefore exactness means that $\widehat{\nu}$ is supported on $L_r\mathbb{Z}$, equivalently that ν is invariant under rotation by $1/L_r$. Fourier uniqueness follows by approximating continuous test functions by trigonometric polynomials. At an irrational M , $H(nM) \neq 0$ for every $n \neq 0$, so exactness even for $j = 0$ forces $\nu = \lambda$.

Editorial note (ProveIt, 2026-09-30). For an integer mesh $M_0 = a$ and the one-atom filter $\nu = \delta_0$, the classification above says that the unfiltered comb is exact through degree r at every phase if and only if $r \leq \nu_2(a)$. This is Theorem `thm:up-composite-mesh` of the canonical synthesis `inverse-and-sampling/comb-interpolation/comb_interpolation_synthesis/` (part *Additive dyadic foundations*, same normalization (2.4)) together with its sharpness, machine-checked in the unscaled form (nodes $k + \theta$, kernel $u(x/M)$) as `Fabius.rvachevCombExactThrough_iff_padicValNat` (Analysis/FabiusFunction/Lean/FabiusFunction/CompositeMeshSharpness.lean). The phase results of that part (`thm:phase-average`, `thm:phase-cubature`, `thm:tm-bernoulli-egf`, `thm:root-unity-filter`, `thm:phase-zero-set`) are exact identities at integer meshes; the synthesis contains no statement at irrational meshes and no approximate, Haar-measure or Wasserstein statement. Theorems 4.1, 5.1, 7.1, 8.1 and 9.1 therefore have no counterpart there.

3 Finite Fourier witnesses: signs versus positivity

Define the finite multiplier barrier

$$B_{n,r}(M) = \max_{0 \leq j \leq r} \frac{|H^{(j)}(nM)|}{(2\pi)^j}, \quad b_{N,r}(M) = \min_{1 \leq n \leq N} B_{n,r}(M). \quad (3.1)$$

The following elementary bounds will be used in two very different limits: $M \rightarrow a/b$ with N fixed, and $N \rightarrow \infty$ with M fixed.

Lemma 3.1 (A mass-one atomic measure has a low Fourier witness). *For every $\nu \in \mathcal{A}_N^{\mathbb{C}}$,*

$$\max_{1 \leq n \leq N} |\widehat{\nu}(n)| \geq \frac{1}{2^N - 1}. \quad (3.2)$$

For every $\nu \in \mathcal{A}_N^+$, the stronger sharp inequality holds:

$$\max_{1 \leq n \leq N} |\widehat{\nu}(n)| \geq \frac{1}{N}. \quad (3.3)$$

Proof. Suppose first that the actual support has $K \leq N$ points. Put $z_h = e^{-2\pi i \phi_h}$ and $A(z) = \prod_{h=1}^K (z - z_h) = \sum_{j=0}^K a_j z^j$. Since A vanishes on the support, $\sum_{j=0}^K a_j \widehat{\nu}(j) = 0$. Here $|a_0| = 1$ and $\widehat{\nu}(0) = 1$. The elementary-symmetric coefficient bound gives $\sum_{j=1}^K |a_j| \leq 2^K - 1$. This proves (3.2)

with K , and hence with N . It uses neither positivity nor a bound on the weights. This is the obstruction already used in [3].

For positive weights w_h , use the Fejér kernel

$$F_N(x) = \sum_{|n| \leq N} \left(1 - \frac{|n|}{N+1}\right) e^{2\pi i n x} = \frac{1}{N+1} \left| \sum_{k=0}^N e^{2\pi i k x} \right|^2. \quad (3.4)$$

It is pointwise nonnegative and $F_N(0) = N+1$. Therefore

$$\begin{aligned} 1 + 2 \sum_{n=1}^N \left(1 - \frac{n}{N+1}\right) |\widehat{v}(n)|^2 &= \sum_{h,k} w_h w_k F_N(\phi_h - \phi_k) \\ &\geq (N+1) \sum_h w_h^2 \geq \frac{N+1}{N}. \end{aligned}$$

Since $2 \sum_{n=1}^N (1 - n/(N+1)) = N$, the largest squared Fourier coefficient is at least $1/N^2$.

Sharpness of (3.3) is witnessed by the equally weighted measure on any N vertices of a regular $(N+1)$ -gon. For $1 \leq n \leq N$, its Fourier coefficient is minus the omitted vertex's n th power divided by N . $\square$

Corollary 3.2 (Explicit error certificates). *For every $M > 0$, $r \geq 0$, and $N \geq 1$,*

$$\mathcal{E}_{N,r}^{\mathbb{C}}(M) \geq \frac{b_{N,r}(M)}{2^N - 1}, \quad \mathcal{E}_{N,r}^+(M) \geq \frac{b_{N,r}(M)}{N}. \quad (3.5)$$

Proof. The absolute value of any Fourier coefficient of D_j is at most its uniform norm. By (2.9), $E_{M,r}(v) \geq b_{N,r}(M) |\widehat{v}(n)|$ for each n . Take the frequency supplied by Lemma 3.1. $\square$

Remark 3.3. The $1/N$ estimate is sharp as a statement about low moments. It does not assert that the resulting quadrature lower bound is sharp in its leading constant. Also, positivity cannot simply be deleted from its proof: the nonnegative off-diagonal Fejér terms are multiplied by products of weights.

4 The exact optimal exponent near every rational mesh

4.1 Statement of the local theorem

Fix $M_0 = a/b > 0$ in lowest terms. Define

$$z_N(a/b) = \begin{cases} 0, & N < b, \\ v_2(a) + 1 + \lfloor \log_2(N/b) \rfloor, & N \geq b, \end{cases} \quad q_{N,r}(a/b) = \max\{0, z_N(a/b) - r\}. \quad (4.1)$$

The first case is essential: a large two-adic numerator does not remove the need to pay for the denominator.

Theorem 4.1 (Sharp rational-resonance exponent). *Let a, b, N be positive integers with $\gcd(a, b) = 1$, and let $r \geq 0$. There are constants $c, C, \eta > 0$, depending on these fixed data, such that for $0 < |M - a/b| < \eta$,*

$$c|M - a/b|^{q_{N,r}(a/b)} \leq \mathcal{E}_{N,r}^{\mathbb{C}}(M) \leq \mathcal{E}_{N,r}^{\mathbb{R}}(M) \leq \mathcal{E}_{N,r}^+(M) \leq C|M - a/b|^{q_{N,r}(a/b)}. \quad (4.2)$$

When the exponent is zero, this means a positive lower bound, not convergence to zero. When it is positive, the explicit positive filter

$$L = b 2^s, \quad s = \lfloor \log_2(N/b) \rfloor, \quad v_L \quad (4.3)$$

attains the order. The lower bound permits arbitrary mesh-dependent nodes and arbitrarily large real or complex weights.

This is a local statement about the optimal *order*, with N and r fixed. It does not state that the same regular polygon always minimizes the leading constant, or that the neighborhood and constants are uniform as the budget grows.

4.2 The finite list that determines the exponent

For $1 \leq n \leq N$ let v_n be the zero order of H at nM_0 , with order zero assigned to a nonzero value. By Lemma 2.1,

$$v_n = \begin{cases} 0, & b \nmid n, \\ d + 1 + v_2(n/b), & b \mid n, \end{cases} \quad d = v_2(a). \quad (4.4)$$

Consequently

$$\max_{1 \leq n \leq N} v_n = z_N(a/b). \quad (4.5)$$

Indeed, for $N \geq b$, the largest two-adic valuation of an integer not exceeding N/b is $\lfloor \log_2(N/b) \rfloor$, attained at a power of two.

Lemma 4.2 (Local multiplier asymptotics). *If $v_n > r$, then, as $\delta \rightarrow 0$,*

$$B_{n,r}(M_0 + \delta) = \frac{|H^{(v_n)}(nM_0)|n^{v_n-r}}{(2\pi)^r(v_n-r)!} |\delta|^{v_n-r} (1 + o(1)). \quad (4.6)$$

If $v_n \leq r$, then $B_{n,r}(M_0) > 0$ and $B_{n,r}$ is continuous there. In particular,

$$b_{N,r}(M_0 + \delta) \asymp |\delta|^{q_{N,r}(M_0)}. \quad (4.7)$$

Proof. Taylor expansion at nM_0 shows that $H^{(j)}(nM_0 + n\delta)$ has order $v_n - j$ when $j < v_n$. If $v_n > r$, the $j = r$ term in the maximum has the lowest order, with the nonzero coefficient in (4.6); all terms with $j < r$ are smaller by a positive power of $|\delta|$. If $v_n \leq r$, the $j = v_n$ term is nonzero at the center. Taking the minimum over a finite list selects its largest vanishing order, giving (4.7). $\square$

For a positive exponent q , the leading constant of $b_{N,r}$ is explicitly

$$\min_{\substack{1 \leq n \leq N \\ v_n = z_N}} \frac{|H^{(z_N)}(nM_0)|n^q}{(2\pi)^r q!}. \quad (4.8)$$

Thus the lower bound is more than an abstract compactness argument: it is controlled by finitely many nonzero half-integer product values through (2.3).

4.3 Proof of the theorem

Proof of Theorem 4.1. The lower bound follows immediately from Corollary 3.2 and Lemma 4.2; the factor $2^N - 1$ is constant in this local limit. The nesting of the three classes gives the middle inequalities.

Suppose first that $q = q_{N,r}(M_0) > 0$. Then $N \geq b$. Let s, L be as in (4.3) and set $p = d + s + 1$, so $q = p - r$. The only nonzero Fourier coefficients of v_L occur at frequencies Ln . At the central mesh,

$$LnM_0 = a2^s n,$$

which is a nonzero integer with zero multiplicity at least p . For $0 \leq j \leq r$ and $0 \leq \ell < q$, one has $j + \ell < p$, hence

$$\left. \frac{\partial^\ell}{\partial M^\ell} H^{(j)}(LnM) \right|_{M=M_0} = (Ln)^\ell H^{(j+\ell)}(LnM_0) = 0.$$

By Lemma 2.2, these are also the derivatives of the uniformly convergent alias series. Taylor's theorem in the uniform norm therefore gives $E_{M_0+\delta,r}(v_L) = O(|\delta|^q)$. This proves the desired upper bound with a positive filter of $L \leq N$ atoms.

If $q = 0$, choose any one-atom filter. Its error is locally bounded as a continuous function of M , by Lemma 2.2. This proves the upper bound with exponent zero. No minimizing signed filter, or convergent subsequence of signed filters, has been used anywhere in the proof. $\square$

Corollary 4.3 (One doubling, one extra robustness order). *Once $N \geq b$ and the requested degree is exactly reproducible at M_0 , the local order increases by one precisely when N crosses a threshold $b2^s$. Increasing r by one decreases that order by one, until the order reaches zero. In particular, allowing signs cannot improve the exponent.*

Example 4.4 (An odd numerator). At $M_0 = 3/5$, constant reproduction has the following optimal local orders:

Phase budget N	Exponent	Consequence
1–4	0	A nonzero error remains at the rational mesh.
5–9	1	Linear robustness.
10–19	2	Quadratic robustness.
20–39	3	Cubic robustness.
40–79	4	Fourth-order robustness.

For degree $r = 2$, the first positive local order occurs at $N = 20$; the orders in the last two rows become one and two. The assertions apply to the best adaptive complex filter as well as to the best positive filter.

Example 4.5 (A numerator with extra two-adic divisibility). At $M_0 = 4/3$, three through five phases give constant error of exact optimal order three, while six through eleven phases give order four. With degree $r = 2$, those orders become one and two. Fewer than three phases cannot make even the constant error vanish near the central mesh.

5 The leading response is a coarse-mesh profile

The upper-bound proof hides a useful explicit formula. Retain $M_0 = a/b$, $a = 2^d m$ with m odd, $L = b2^s$, $p = d + s + 1$, and assume $r < p$. Set $q = p - r$ and define the smooth periodic function

$$\Psi_{m,r}(x) = \sum_{\substack{n \neq 0 \\ n \text{ odd}}} \left(\frac{i}{2\pi n} \right)^r H(mn/2) e^{2\pi i n x}. \quad (5.1)$$

Rapid decrease makes this series and all its derivatives uniformly convergent. It is real valued, and its first Fourier coefficient is nonzero.

Theorem 5.1 (Explicit leading profile). *Uniformly in θ , as $\delta \rightarrow 0$,*

$$D_r(M_0 + \delta, \nu_L; \theta) = -\frac{p!}{q! L^r M_0^p} \delta^q \Psi_{m,r}(L\theta) + O(|\delta|^{q+1}), \quad (5.2)$$

$$D_j(M_0 + \delta, \nu_L; \theta) = O(|\delta|^{p-j}) \quad (0 \leq j < r). \quad (5.3)$$

Consequently

$$E_{M_0+\delta,r}(\nu_L) = \frac{p! \|\Psi_{m,r}\|_\infty}{q! L^r M_0^p} |\delta|^q + O(|\delta|^{q+1}), \quad (5.4)$$

with a strictly positive leading constant.

Proof. In the alias expansion for D_r , the coefficient of δ^q at frequency Ln is

$$\left(\frac{i}{2\pi} \right)^r \frac{(Ln)^q}{q!} H^{(p)}(a2^s n).$$

It is zero when n is even, because the multiplicity then exceeds p . For odd n , Lemma 2.1 makes it

$$-\frac{p!}{q! L^r M_0^p} \left(\frac{i}{2\pi n} \right)^r H(mn/2).$$

This proves (5.2); the uniform remainder follows from the next uniform M -derivative in Lemma 2.2. The same argument through degree $p - j - 1$ proves (5.3). The uniform norm is Lipschitz with respect to uniform perturbations, so taking the maximum over j gives (5.4). Nonvanishing of the first coefficient implies $\|\Psi_{m,r}\|_\infty > 0$. $\square$

For constants, the profile has a particularly direct interpretation:

$$\Psi_{m,0}(x) = Q_{m/2,x}(1) - 1. \quad (5.5)$$

Indeed the even aliases at the mesh $m/2$ are $H(mk) = 0$ for $k \neq 0$. Thus the first nonzero response near a large refined rational mesh is a rescaled defect of the fixed half-integer mesh determined by the odd part of the numerator. In particular,

$$D_0(M_0 + \delta, \nu_L; \theta) = -\left(\frac{\delta}{M_0}\right)^p (Q_{m/2,L\theta}(1) - 1) + O(|\delta|^{p+1}). \quad (5.6)$$

When $m = 1$, this yields an exact leading norm constant:

$$E_{M_0+\delta,0}(\nu_L) = \left|\frac{\delta}{M_0}\right|^p + O(|\delta|^{p+1}). \quad (5.7)$$

To see this without numerical evaluation, write $Y = V_1/2 + Z$ with $|Z| \leq 1/2$. Convolution with the uniform density on $[-1/2, 1/2]$ gives $0 \leq u \leq 1$ and $u(0) = 1$. At mesh $1/2$, at most one nonzero interior sample occurs, so $0 \leq Q_{1/2,x}(1) \leq 2$. Its values at $x = 0$ and $x = 1/2$ are two and zero, respectively. Hence $\|\Psi_{1,0}\|_\infty = 1$.

Remark 5.2 (What is and is not optimized). Equation (5.4) is an asymptotic equivalence for the specified regular polygon. Theorem 4.1 optimizes its exponent over all allowed filters. The leading constant for the fully optimized problem may be smaller; we do not identify it here.

6 Two-sided Fourier estimates with explicit envelopes

6.1 A uniform upper envelope for every fixed derivative

For $t \geq 1$ put

$$G(t) = t^{-1/2} \exp\left(-\frac{(\log t)^2}{2 \log 2}\right). \quad (6.1)$$

Lemma 6.1 (Differentiated log-square envelope). *For every integer $r \geq 0$ and real $|t| \geq 1$,*

$$\max_{0 \leq j \leq r} \frac{|H^{(j)}(t)|}{(2\pi)^j} \leq 2^r G(|t|). \quad (6.2)$$

Proof. It suffices to consider $t \geq 1$. Write $f(x) = \sin(\pi x)/(\pi x)$. For $|x| \geq 1$, $|f(x)| \leq 1/|x|$. For $\ell \geq 1$, differentiating $f(x) = \frac{1}{2} \int_{-1}^1 e^{i\pi xs} ds$ and integrating by parts once shows

$$|f^{(\ell)}(x)| \leq \frac{2\pi^{\ell-1}}{|x|} \leq \frac{(2\pi)^\ell}{|x|}.$$

Let $J = \lfloor \log_2 t \rfloor + 1$ and split $H(t) = \prod_{h=0}^{J-1} f(t/2^h) H(t/2^J)$. Each initial argument is at least one. Also, because $|Y| \leq 1$, $|H^{(\ell)}(y)| \leq (2\pi)^\ell$ for every real y . Leibniz's rule therefore gives

$$|H^{(j)}(t)| \leq (2\pi)^j \left(\sum_{h=0}^J 2^{-h}\right)^j \frac{2^{J(J-1)/2}}{t^J} \leq (4\pi)^j \frac{2^{J(J-1)/2}}{t^J}.$$

Writing $\log_2 t = k + \xi$, $0 \leq \xi < 1$, shows

$$\log \left(\frac{2^{J(J-1)/2}}{t^J} \right) = -\frac{(\log t)^2}{2 \log 2} - \frac{1}{2} \log t + \frac{\log 2}{2} (\xi^2 - \xi) \leq \log G(t).$$

Divide by $(2\pi)^J$ and take the maximum. $\square$

Corollary 6.2 (An explicit regular-grid error bound). *For $MN \geq 2$,*

$$E_{M,r}(v_N) \leq 6 \cdot 2^r G(MN). \quad (6.3)$$

Proof. Only aliases Nn survive. By Lemma 6.1, their absolute sum is at most $2^{r+1} \sum_{n \geq 1} G(MNn)$. Moreover,

$$\frac{G(xn)}{G(x)} = n^{-1/2 - \log_2 x} \exp \left(-\frac{(\log n)^2}{2 \log 2} \right) \leq n^{-3/2} \quad (x \geq 2).$$

The elementary integral bound gives $\sum_{n \geq 1} n^{-3/2} \leq 3$. $\square$

6.2 A lower envelope away from arithmetic resonances

Write $\|t\|_{\mathbb{Z}} = \text{dist}(t, \mathbb{Z})$. The lower bound must retain this distance: a global positive lower envelope for H is impossible because H has zeros.

Lemma 6.3 (A nonzero product lower bound). *If $t \geq 1$ is not an integer and $J = \lceil \log_2(2t) \rceil$, then*

$$|H(t)| \geq c_* \left(\frac{2\|t\|_{\mathbb{Z}}}{\pi t} \right)^J, \quad c_* = e^{-\pi^2/9}. \quad (6.4)$$

More generally, for $0 \leq s \leq 1/2$,

$$e^{-4\pi^2 s^2/9} \leq H(s) \leq 1. \quad (6.5)$$

Proof. For every real x , $|\sin(\pi x)| \geq 2\|x\|_{\mathbb{Z}}$. Since $2^h \mathbb{Z} \subseteq \mathbb{Z}$,

$$\left\| \frac{t}{2^h} \right\|_{\mathbb{Z}} = 2^{-h} \text{dist}(t, 2^h \mathbb{Z}) \geq 2^{-h} \|t\|_{\mathbb{Z}}.$$

Thus each of the first J sinc factors has magnitude at least $2\|t\|_{\mathbb{Z}}/(\pi t)$.

For the tail let $x_h = \pi s/2^h$. Taylor's elementary lower bound gives $\sin x_h/x_h \geq 1 - x_h^2/6$. Here $x_h^2/6 \leq \pi^2/24 < 1/2$, and $\log(1 - y) \geq -2y$ on $[0, 1/2]$. Summing gives

$$\log H(s) \geq -\frac{1}{3} \sum_{h \geq 0} x_h^2 = -\frac{4\pi^2 s^2}{9}.$$

The upper bound is immediate. Since $t/2^J \leq 1/2$, this proves (6.4) as well. $\square$

Definition 6.4 (Finite-type mesh). An irrational $M > 0$ satisfies a finite-type bound if there are constants $\kappa > 0$ and $\tau \geq 1$ such that

$$\|nM\|_{\mathbb{Z}} \geq \kappa n^{-\tau} \quad (n \geq 1). \quad (6.6)$$

The constants are part of the hypothesis. We do not assume a uniform κ over all meshes.

Proposition 6.5 (A finite-type spectral floor). *Under (6.6), there exist constants $B, C > 0$ such that for every $N \geq 2$,*

$$h_M(N) := \min_{1 \leq n \leq N} |H(nM)| \geq \exp \left[-\frac{\tau+1}{\log 2} (\log N)^2 - B \log N - C \right]. \quad (6.7)$$

Proof. For $nM \geq 1$, use Lemma 6.3. Set $a_0 = \min\{1, 2\kappa/(\pi M)\} > 0$ and choose a fixed $C_M \geq 1$ such that $\lceil \log_2(2nM) \rceil \leq \log_2 n + C_M$ for all such n . Then

$$|H(nM)| \geq c_* (a_0 n^{-(\tau+1)})^{\log_2 n + C_M}.$$

Taking logarithms gives the claimed quadratic leading coefficient and linear remainder. The finitely many indices with $nM < 1$ have nonzero values; enlarge C to include their minimum. Finally replace n by the upper bound N in the resulting decreasing lower envelope. $\square$

Example 6.6 (A fully elementary finite-type example). For $M = \sqrt{2}$, let k be the nearest integer to $n\sqrt{2}$. The nonzero integer $|2n^2 - k^2|$ is at least one. Also $n\sqrt{2} + k < 4n$ for $n \geq 1$. Hence

$$\|n\sqrt{2}\|_{\mathbb{Z}} = \frac{|2n^2 - k^2|}{n\sqrt{2} + k} \geq \frac{1}{4n}.$$

Thus (6.6) holds with $\kappa = 1/4$ and $\tau = 1$, without appealing to a deeper approximation theorem.

7 Optimal positive phase complexity at irrational meshes

Theorem 7.1 (The log-square phase-cost scale). *Fix $r \geq 0$ and an irrational mesh M satisfying (6.6). Then*

$$\mathcal{E}_{N,r}^+(M) = \exp[-\Theta((\log N)^2)] \quad (N \rightarrow \infty). \quad (7.1)$$

More explicitly,

$$\frac{1}{2 \log 2} \leq \liminf_{N \rightarrow \infty} \frac{-\log \mathcal{E}_{N,r}^+(M)}{(\log N)^2} \leq \limsup_{N \rightarrow \infty} \frac{-\log \mathcal{E}_{N,r}^+(M)}{(\log N)^2} \leq \frac{\tau + 1}{\log 2}. \quad (7.2)$$

If $N_{M,r}(\varepsilon)$ is the smallest positive phase budget whose optimal error is at most ε , then

$$N_{M,r}(\varepsilon) = \exp[\Theta(\sqrt{\log(1/\varepsilon)})] \quad (\varepsilon \downarrow 0). \quad (7.3)$$

Proof. By Corollary 3.2, using just the zeroth multiplier,

$$\frac{h_M(N)}{N} \leq \mathcal{E}_{N,r}^+(M) \leq E_{M,r}(\nu_N) \leq 6 \cdot 2^r G(MN) \quad (7.4)$$

once $MN \geq 2$. The lower estimate (6.7) and the explicit formula for G prove (7.1) and (7.2).

The positive infimum is attained: represent a filter by N circle points and a point of the compact probability simplex, allowing repeated nodes and zero weights. The error is continuous in these parameters, uniformly in θ . Thus there is a minimum. The optimized error is nonincreasing in N , is positive for each finite N by (7.4), and tends to zero by the grid upper bound. Inverting the two log-square inequalities proves (7.3). $\square$

7.1 Interpretation and limitations

The result says that regular polygons have the correct *logarithmic scale* of complexity among all positive filters, even those with highly nonuniform weights and nonuniformly spaced nodes. It does not say they have the smallest error at each N , nor that the lower and upper constants in (7.2) coincide. For the elementary mesh $M = \sqrt{2}$, those constants are $1/(2 \log 2)$ and $2/\log 2$.

The distinction from signed filters is real. For unrestricted complex weights, our unconditional lower estimate is only

$$\mathcal{E}_{N,r}^{\mathbb{C}}(M) \geq \frac{h_M(N)}{2^N - 1}. \quad (7.5)$$

This is enough to prove the exact fixed- N rational-resonance exponent, but not enough to prove (7.1) when N grows. We do not replace $2^N - 1$ by N for signed weights, and do not claim the positive complexity law for that larger class.

The theorem also must not be confused with a claim about arbitrary Gaussian quadrature. The rule family retains its kernel-dependent physical weights, and one filter must work for every overall translation. Removing that translation quantifier changes the optimization problem fundamentally.

8 Approximate Haar rigidity and a finite-frequency inverse bound

For probability measures on the circle, define the Wasserstein distance by its usual dual formula

$$W_1(\mu, \nu) = \sup_{\text{Lip}(f) \leq 1} \left| \int_{\mathbb{R}} f d(\mu - \nu) \right|, \quad (8.1)$$

where the metric on $\mathbb{R}/\mathbb{Z}$ is the geodesic distance of circumference one. For fixed irrational M and degree r , put

$$\omega_{M,r}(\varepsilon) = \sup\{W_1(\mu, \lambda) : \mu \text{ a probability measure on } \mathbb{R}/\mathbb{Z}, E_{M,r}(\mu) \leq \varepsilon\}. \quad (8.2)$$

The measure in this definition need not be atomic.

Theorem 8.1 (Finite-frequency inversion). *For irrational $M > 0$, every probability measure μ , and every integer $K \geq 2$,*

$$W_1(\mu, \lambda) \leq \frac{8}{K} + \frac{E_{M,0}(\mu)}{\pi} \sum_{n=1}^K \frac{1}{n|H(nM)|} \leq \frac{8}{K} + \frac{E_{M,0}(\mu)(1 + \log K)}{\pi h_M(K)}. \quad (8.3)$$

In particular, $\omega_{M,r}(\varepsilon) \rightarrow 0$ as $\varepsilon \downarrow 0$ for every fixed irrational mesh, without a finite-type assumption.

Proof. Let $m = \lfloor K/2 \rfloor$ and $L = m + 1$. Normalize the square of the Fejér kernel to obtain the nonnegative trigonometric polynomial

$$J_m(x) = \frac{F_m(x)^2}{A_m}, \quad A_m = \int_{\mathbb{R}} f_m^2 d\lambda = \frac{2L^2 + 1}{3L}.$$

Its degree is $2m \leq K$ and its integral is one. On $|x| \leq 1/2$, $F_m(x) \leq L$ and $F_m(x) \leq 1/(4Lx^2)$. Since $A_m \geq 2L/3$, splitting the first absolute moment at $1/L$ gives

$$\begin{aligned} \int_{\mathbb{R}} |x| J_m(x) dx &\leq 2 \int_0^{1/L} x \frac{3L}{2} dx + 2 \int_{1/L}^{1/2} x \frac{3}{32L^3 x^4} dx \\ &\leq \frac{3}{2L} + \frac{3}{32L} < \frac{2}{L}. \end{aligned}$$

Thus every one-Lipschitz function satisfies $\|f - f * J_m\|_{\infty} \leq 2/L$.

A periodic Lipschitz function is absolutely continuous, and integration by parts gives $|\widehat{f}(n)| \leq 1/(2\pi|n|)$ for $n \neq 0$. Also $|\widehat{J}_m(n)| \leq 1$. With $\sigma = \mu - \lambda$, whose total variation is at most two, we obtain

$$\left| \int f d\sigma \right| \leq \frac{4}{L} + \frac{1}{\pi} \sum_{n=1}^K \frac{|\widehat{\mu}(n)|}{n}.$$

The zeroth-moment alias identity implies $|\widehat{\mu}(n)| \leq E_{M,0}(\mu)/|H(nM)|$. Finally $4/L \leq 8/K$ and $\sum_{n=1}^K 1/n \leq 1 + \log K$.

For the convergence assertion, first choose a fixed K making $8/K$ small. Every factor in $h_M(K)$ is nonzero. Then choose ε small enough for the second term, using $E_{M,0} \leq E_{M,r}$. $\square$

Lemma 8.2 (Distance of a regular polygon from Haar measure). *For every $N \geq 1$,*

$$W_1(\nu_N, \lambda) = \frac{1}{4N}. \quad (8.4)$$

Proof. Partition the circle into the N equal cells centered at the grid nodes. Sending each cell to its center bounds every Lipschitz test difference by $N \int_{-1/(2N)}^{1/(2N)} |x| dx = 1/(4N)$. For the reverse inequality use the one-Lipschitz function equal to distance from the grid. It is zero at every atom and has Haar integral $1/(4N)$. $\square$

Theorem 8.3 (The inverse scale and failure of every Hölder estimate). *Fix $r \geq 0$. For every irrational $M > 0$, no constants $C, \alpha > 0$ give a local estimate*

$$W_1(\mu, \lambda) \leq C E_{M,r}(\mu)^\alpha \quad (8.5)$$

near Haar measure. If M satisfies (6.6), then

$$\omega_{M,r}(\varepsilon) = \exp[-\Theta(\sqrt{\log(1/\varepsilon)})]. \quad (8.6)$$

For clarity about constants, if

$$0 < \alpha_0 < \sqrt{\frac{\log 2}{\tau + 1}}, \quad \beta_0 > \sqrt{2 \log 2},$$

then, for all sufficiently small ε ,

$$e^{-\beta_0 \sqrt{\log(1/\varepsilon)}} \leq \omega_{M,r}(\varepsilon) \leq e^{-\alpha_0 \sqrt{\log(1/\varepsilon)}}. \quad (8.7)$$

Proof. For the non-Hölder assertion, use $\mu = \nu_N$. By Corollary 6.2 and Lemma 8.2, its error is at most $6 \cdot 2^r G(MN)$, while its distance is $1/(4N)$. For every fixed $\alpha > 0$ the ratio of this distance to $[6 \cdot 2^r G(MN)]^\alpha$ tends to infinity: the positive quadratic term in $\log N$ dominates every linear term. Meanwhile $\nu_N \rightarrow \lambda$ in W_1 , so these counterexamples occur arbitrarily close to Haar measure.

For the finite-type upper bound write $T = \log(1/\varepsilon)$ and choose α_1 strictly between α_0 and $\sqrt{\log 2/(\tau + 1)}$. Put $K = \lfloor e^{\alpha_1 \sqrt{T}} \rfloor$. Substitute (6.7) into (8.3). Its second term is at most

$$\exp \left[- \left(1 - \frac{\tau + 1}{\log 2} \alpha_1^2 \right) T + O(\sqrt{T} + \log T) \right],$$

which is smaller than the first term on the stated scale. The slack between α_0 and α_1 absorbs the fixed prefactors.

For the lower bound no arithmetic assumption is needed. Choose

$$N = \left\lceil M^{-1} \exp \left(\sqrt{2 \log 2} [T + \log(6 \cdot 2^r)] \right) \right\rceil.$$

For large T , Corollary 6.2 gives $E_{M,r}(\nu_N) \leq \varepsilon$. Equation (8.4) then gives $\log \omega_{M,r}(\varepsilon) \geq -\sqrt{2 \log 2} \sqrt{T} + O(1)$. The slack in β_0 proves the lower inequality. $\square$

Remark 8.4 (A deterministic statement, not a statistical rate). The error is a uniform deterministic quadrature defect, and the measure being recovered is the phase distribution. There are no random observations or sample-size assumptions in Theorems 8.1 and 8.3. Statistical Wasserstein deconvolution results such as [2] provide context, not a black-box proof of these particular rates.

9 Liouville meshes: arbitrarily fast approximation and slow recovery

Exact irrational rigidity alone imposes no mesh-independent lower scale for finite positive approximation. The following theorem makes this failure quantitative and keeps the mesh fixed after its construction.

Theorem 9.1 (Arbitrary-rate obstruction). *Fix $r \geq 0$, a nonempty open interval $I \subset (0, \infty)$, and any function $\rho : (0, 1) \rightarrow (0, \infty)$ with $\rho(\varepsilon) \rightarrow 0$ as $\varepsilon \downarrow 0$. There exist a Liouville irrational $M \in I$, integers $N_j \rightarrow \infty$, and $\varepsilon_j \downarrow 0$ such that*

$$E_{M,r}(v_{N_j}) \leq \varepsilon_j, \quad W_1(v_{N_j}, \lambda) = \frac{1}{4N_j} > \rho(\varepsilon_j). \quad (9.1)$$

Consequently $\omega_{M,r}(\varepsilon_j) > \rho(\varepsilon_j)$. Furthermore, for any prescribed positive sequence (η_N) , the construction can require $\varepsilon_j \leq \eta_{N_j}$.

Proof. Enumerate the positive rationals as $q_1, q_2, \dots$. Start with a closed interval of positive length contained in I . Inductively choose a reduced rational a_j/b_j in the interior of the previous interval, different from $q_1, \dots, q_j$, with b_j increasing without bound. Such rationals are dense. Put $N_j = 2^r b_j$, choosing the denominators large enough that N_j is strictly increasing.

At the mesh a_j/b_j , the regular N_j -gon refines to the integer mesh $2^r a_j$. Its integer aliases have multiplicity at least $r + 1$, so its error through degree r is zero. Choose

$$0 < \varepsilon_j < \min\{2^{-j}, \varepsilon_{j-1}/2, \eta_{N_j}\}$$

with the last condition omitted when no sequence is prescribed, and with $\rho(\varepsilon_j) < 1/(4N_j)$. The limit assumption on ρ permits this. For the first step omit $\varepsilon_{j-1}/2$.

Continuity of the fixed-grid error in the mesh gives a neighborhood of a_j/b_j throughout which that error is less than ε_j . Choose a closed interval of positive length centered there, contained in this neighborhood and the interior of the previous interval, excluding $q_1, \dots, q_j$, of diameter at most 2^{-j} , and of radius less than b_j^{-j} . Every one of these conditions allows a positive radius.

The nested intervals have a unique common point M . Exclusion of the rationals makes M irrational. Also $0 < |M - a_j/b_j| < b_j^{-j}$, so M is Liouville in the usual approximation sense. The error inequality holds for every j because the final point belongs to the j th interval. The distance assertion is Lemma 8.2, and the chosen inequality for ρ completes the proof. $\square$

The two conclusions describe the same phenomenon from opposite directions. Exceptionally accurate rational approximations make a finite phase grid exceptionally effective. But precisely because it remains a finite grid, its distance from Haar measure is still $1/(4N)$; a tiny defect can therefore carry very little information about the phase distribution.

There is no contradiction with Theorem 8.1: for each constructed irrational M , its inverse modulus still tends to zero. Nor does the theorem claim that every Liouville mesh exhibits the same behavior. It constructs meshes tailored to the prescribed obstruction.

10 Reproducibility, finite certificates, and implementation

10.1 What can be checked exactly

The exponent in Theorem 4.1 requires only integer arithmetic. For reduced a/b , compute the orders in (4.4) for $1 \leq n \leq N$, take their maximum, and subtract r with truncation at zero. An equivalent constant-time arithmetic formula after integer division is (4.1); $\lfloor \log_2 \lfloor N/b \rfloor \rfloor$ can be computed from the bit length of the positive integer $\lfloor N/b \rfloor$, without floating-point logarithms.

The supplied program compares these two routes over a finite parameter grid. It also verifies that positivity of the exponent is exactly the inherited support threshold (1.1), and that the proposed regular polygon never exceeds the budget. These checks are finite regression tests; the general statements are proved in the article.

The Fejér coefficient sum and its squared-kernel normalization are checked with rational arithmetic. Numerical tests of the positive and complex Fourier witnesses supplement, but do not replace, their proofs. In particular, no sampled collection of positive weights establishes the universal $1/N$ inequality by itself.

10.2 A product enclosure usable by a certified implementation

For real t and a truncation index J with $|t|/2^J \leq 1/2$, define

$$P_J(t) = \prod_{h=0}^{J-1} \frac{\sin(\pi t/2^h)}{\pi t/2^h}.$$

The tail is positive. Lemma 6.3 gives the analytic enclosure

$$|P_J(t)| \exp\left(-\frac{4\pi^2 t^2}{9 \cdot 4^J}\right) \leq |H(t)| \leq |P_J(t)|. \quad (10.1)$$

Thus the infinite tail does not require an uncontrolled numerical cutoff. A fully certified implementation would evaluate the finite product using outward-rounded interval arithmetic and combine it with (10.1). Derivative enclosures can instead be obtained by finite Taylor jets and differentiated tails.

The bundled numerical code uses high-precision `mpmath`, not directed-rounding interval arithmetic. Its evaluations are diagnostics, not certified enclosures of the exact values. It uses the displayed analytic tail estimate to make truncation negligible compared with the test tolerance. The distinction between an analytic enclosure formula and a floating-point implementation of its finite part is important.

10.3 Recorded checks and limits

The script records its exact assertion count, random seed, numerical precision, parameter ranges, and residuals in `validation/results.json`. It checks the Fourier upper and lower envelopes at nonintegral test points, the leading integer-zero formula, the predicted local powers for regular grids, and the sharp low-moment examples. A human-readable transcript is included. The delivered run passed 358,664 exact assertions and 1,465 numerical diagnostics at 90 decimal digits. The largest relative residual in the leading-zero-coefficient tests was approximately 2.29×10^{-12} , using displacements of magnitude 10^{-12} . The source contains no network calls.

The local diagnostic tracks individual alias coefficients near rational meshes; where a full defect is sampled, the output is explicitly labeled as a sampled trigonometric approximation. A sampled maximum is not reported as the true continuous supremum. No numerical result is used as a premise of a theorem above.

11 A formalization route and a dependency audit

The article does not ship an untested Lean file labeled as a formal proof. Instead, the following decomposition identifies concrete interfaces for a future formalization. The repository already has the Rvachev synthesis and alias-analysis development [4, 5]; its exact current interfaces must be inspected before any reuse.

Layer	Required statement or construction
Finite arithmetic	Reduced-fraction divisibility; maximum two-adic valuation below an integer bound; the exponent formula and support threshold.
Atomic moments	The annihilating polynomial identity for complex weights; Fejér nonnegativity and the positive $1/N$ witness.
Kernel jets	The exact zero multiplicity and leading derivative of the infinite product; smooth parameter dependence of the alias series.
Local theorem	Uniform Taylor expansion; a finite minimum of functions with known positive leading coefficients; regular-grid refinement.
Large-budget bounds	The explicit differentiated envelope and the finite-type product floor, followed by the optimized phase-cost inequalities.
Inverse theorem	The normalized squared Fejér kernel, its first absolute moment bound, and the dual Wasserstein estimate.
Arithmetic pathology	Nested rational intervals avoiding an enumeration, continuity of a fixed-grid error, and the Liouville approximation property.

The principal logical dependencies are short. The local optimal exponent uses only kernel zero orders, the alias identity, a finite Fourier witness, and Taylor’s theorem. It does not depend on the finite-type hypothesis, the Wasserstein theorem, or the Liouville construction. The positive log-square law uses the stronger positive witness and the two envelope lemmas; it does not depend on a claim of optimal local leading constants. The inverse theorem uses nonzero Fourier multipliers and smoothing; its qualitative version does not require any Diophantine inequality.

These separations are useful both mathematically and for auditing a proof assistant implementation. In particular, no compactness of unbounded signed measures is silently assumed. The only optimization existence statement is for positive finite-support measures, where the simplex parameterization is compact.

12 Further research questions and topics

The questions below are proposed next projects, not assertions that every formulation is globally open. Each needs a separate literature and repository check before a priority claim.

Question 12.1 (Optimal leading constants at a rational resonance). For fixed $a/b, N, r$ with positive exponent q , does $\mathcal{E}_{N,r}^{\mathbb{K}}(M)/|M - a/b|^q$ have a limit as $M \rightarrow a/b$ from each side? Determine it for $\mathbb{K} = +, \mathbb{R}, \mathbb{C}$.

Theorem 4.1 gives two-sided bounds and Theorem 5.1 gives the constant of a regular polygon. The remaining issue is a singular optimization over moving nodes and weights, not nonvanishing of the exponent. Small cases such as $M_0 = 1, N = 1, 2, 3$, and $r = 0$ are concrete starting points.

Question 12.2 (Uniform rational neighborhoods as the budget grows). Find useful bounds on the constants and the radius in Theorem 4.1 when N, a , or b varies. Describe the crossover when $|M - a/b|$ and N tend to their limits together.

Nearby rational resonances can then compete. A fixed finite list of aliases is no longer enough to obtain uniform information without additional estimates.

Question 12.3 (The leading log-square constant). Does the quotient in (7.2) converge for natural fixed meshes, for example $M = \sqrt{2}$? Determine the optimal constant or its liminf and limsup. Are regular polygons asymptotically optimal beyond the scale?

The present constants leave a genuine gap. The arithmetic of the entire list $H(M), \dots, H(NM)$ and the restrictions among a positive measure’s Fourier coefficients both matter; a minimum-multiplier bound may lose information.

Question 12.4 (Signed filters at a growing budget). Determine the asymptotic size of $\mathcal{E}_{N,r}^{\mathbb{R}}(M)$ and $\mathcal{E}_{N,r}^{\mathbb{C}}(M)$ at finite-type irrational meshes. What changes under a specified total-variation bound, allowed to depend on N ?

The exact local exponent cannot improve by using signs, but that does not settle the large- N problem. The gap between $1/N$ and $1/(2^N - 1)$ in the available Fourier witnesses identifies the obstruction in the present proof.

Question 12.5 (Necessary arithmetic conditions). Characterize those irrational meshes for which the log-square complexity law, or the stretched-logarithmic inverse scale, holds. Is a condition on a weighted sequence of rational approximations both necessary and sufficient?

Finite type is sufficient and tailored Liouville examples show that some restriction is necessary for a universal theorem. Neither statement proves that finite type is the exact boundary.

Question 12.6 (Sharper inverse constants and other metrics). Find the optimal modulus of Haar recovery, rather than its scale. Compare W_p , discrepancy, negative Sobolev norms, and recovery under smoothness or separation assumptions on the unknown phase measure.

The regular-grid counterexamples rule out Hölder recovery in W_1 with no extra assumptions. They do not settle every metric or every restricted model.

Question 12.7 (Several dimensions and correlated filters). For a tensor-product Rvachev kernel, determine local resonance exponents and large-budget costs for correlated phase measures in several dimensions.

A zero in one coordinate can annihilate a multidimensional alias. Therefore a direct product of the one-dimensional support thresholds need not be a necessary bound. The finite-frequency witness should be adapted to that geometry rather than assumed to tensor automatically.

Question 12.8 (Other refinement bases and other kernels). Which parts of the local theorem extend to smooth kernels with explicitly known zero orders, and what replaces the log-square scale for products with other contraction schedules?

The lower-bound mechanism is general: a finite moment witness converts multiplier orders into an optimal obstruction. A matching upper construction requires a compatible alias set, which is an additional structural condition.

Question 12.9 (Certified finite-frequency implementation). Implement interval-certified values of $b_{N,r}(M)$ and $h_M(K)$, return a verifiable error or Wasserstein certificate, and formalize the finite arithmetic and moment-witness layers first.

The formula (10.1) supplies an explicit infinite-tail bound. The remaining work includes robust finite-product evaluation near multiple zeros, derivative jets, and a representation of the mesh that supports the required arithmetic nonvanishing certificates.

13 Conclusion

The exact symmetry classification of phase averaging has a quantitative extension, but not a uniformly well-conditioned one. At a fixed rational resonance, the optimal local power is completely determined by the denominator, the numerator's two-adic valuation, the phase budget, and the polynomial degree. Adaptive signed or complex weights cannot improve that exponent. An explicit regular polygon realizes it, with a leading response inherited from a coarse half-integer mesh.

At a fixed finite-type irrational mesh, positivity yields a much stronger large-budget statement: optimal phase errors have log-square decay, and the corresponding phase cost is exponential in the square root of the logarithm of the requested accuracy. The same scale governs approximate Haar recovery. No Hölder inverse bound can hold, even locally at Haar measure. Without arithmetic restrictions, specially constructed irrational meshes make both approximation and inverse instability as extreme along subsequences as a prescribed rate demands.

The remaining tasks are now more specific than the original stability question: optimal leading constants, uniform crossover laws, signed large-budget complexity, and the exact arithmetic boundary. The proofs and reproducibility package separate these unresolved refinements from the results established here.

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